

Assignment 2

February 20, 2026

Question 1 (25 Points). *Three airports in New York City run 24 hours a day, 7 days a week. In a given day, there are requirements for the total number of air traffic controllers that must be at the airports. These are given in Table 1.*

Air traffic controllers can either work 8-hour or 12-hour shifts, starting at the times stated in Table 1 (12-hour shifts can start only at 12 a.m./p.m. or 8 a.m./p.m.). Those working 8-hour shifts cost \$93/h in salary and benefits, and those working 12-hour shifts cost \$100.8/h.

- 1. Formulate and solve a linear program to minimize the dispatcher labor costs. [10 pts]*
- 2. Suppose at most one-third of its controllers can work 12-hour shifts. Repeat (1). You may ignore the requirement that the number of employees must be an integer. [7 pts]*
- 3. What changes would you make to your model to staff the three airports separately? What additional data would you need to request to implement this change? [8 pts]*

Hours	Controllers needed
12 a.m. to 4 a.m.	8
4 a.m. to 8 a.m.	10
8 a.m. to 12 p.m.	16
12 p.m. to 4 p.m.	21
4 p.m. to 8 p.m.	18
8 p.m. to 12 a.m.	12

Table 1: Number of controllers needed during different time intervals.

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Correct Answer 1. 1) Let

x_t = number of controllers starting 8-hours shifts at $t = 0, 4, 8, 12, 16, 20$

y_t = number of controllers starting 12-hours shifts at $t = 0, 8, 12, 20$

$$\min 8(93)(x_0+x_4+x_8+x_{12}+x_{16}+x_{20})+ 12(100.8)(y_0+y_8+y_{12}+y_{20})$$

$$\begin{array}{llllll} s.t. & x_0 & & +x_{20} & + & y_0 & +y_{20} & \geq 8 & (12a.m.-4a.m.) \\ & x_0+x_4 & & & + & y_0 & +y_{20} & \geq 10 & (4a.m.-8a.m.) \\ & & x_4+x_8 & & + & y_0+y_8 & & \geq 16 & (8a.m.-12p.m.) \\ & & & x_8+x_{12} & + & y_8+y_{12} & & \geq 21 & (12p.m.-4p.m.) \\ & & & & x_{12}+x_{16} & + & y_8+y_{12} & \geq 18 & (4p.m.-8p.m.) \\ & & & & & x_{16}+x_{20} & + & y_{12}+y_{20} & \geq 12 & (8p.m.-12a.m.) \\ & & & & & & & & x_t, y_t \geq 0. \end{array}$$

Optimal cost = \$31,713.60

8-hour shift begins	Number of employees
12a.m.	8
4a.m.	2
8a.m.	14
12p.m.	6
4p.m.	11
12-hour shift begins	Number of employees
12p.m.	1

2) Add the constraint:

$$y_0 + y_8 + y_{12} + y_{20} \leq \frac{1}{3}(x_0 + x_4 + x_8 + x_{12} + x_{16} + x_{20} + y_0 + y_8 + y_{12} + y_{20})$$

The optimal solution in part (1) already satisfies the new constraint, so it remains optimal.

3) Index airports by $a \in \{1, 2, 3\}$. To staff the three airports separately, we encode the decision variables per airport. That is, we let

$x_{a,t}$ = number of controllers starting 8-hours shifts at $t = 0, 4, 8, 12, 16, 20$ at airport a

$y_{a,t}$ = number of controllers starting 12-hours shifts at $t = 0, 8, 12, 20$ at airport a

To set up the constraints, we would need to use the number of controllers $d_{a,t}$ needed during the time interval starting at t at every airport a . This is the additional data needed to implement this change. Using these values, we write the demand constraints separately for each airport, and minimize the total cost (assuming that controllers costs are the same in all airports):

$$\min \sum_a \left((8)(93) \sum_{t \in \{0,4,8,12,16,20\}} x_{a,t} + (12)(100.8) \sum_{t \in \{0,8,12,20\}} y_{a,t} \right)$$

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Question 2 (20 Points). Consider the LP below and write out its dual [8 pts]. Solve both the primal and dual (by hand or using a solver of your choice) [6 pts].

Briefly explain how your solutions satisfy the duality relationships: weak duality, strong duality, and complementary slackness [6 pts].

$$\begin{aligned} \max Z = & 10x_1 + 14x_2 + 20x_3 \\ \text{s.t. } & 2x_1 + 3x_2 + 4x_3 \leq 220 \\ & 4x_1 + 2x_2 - x_3 \leq 385 \\ & x_1 + 4x_3 \leq 160 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Correct Answer 2. Dual problem:

$$\begin{aligned} \min W = & 220y_1 + 385y_2 + 160y_3 \\ \text{s.t. } & 2y_1 + 4y_2 + y_3 \geq 10 \\ & 3y_1 + 2y_2 \geq 14 \\ & 4y_1 - y_2 + 4y_3 \geq 20 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

One optimal solution is $x = (60, 0, 25)$ and $y = (5, 0, 0)$, and the optimal value is 1100.

```
from gurobipy import GRB, Model

m1 = Model("Primal_LP")

x1 = m1.addVar(vtype="C", name="x1")
x2 = m1.addVar(vtype="C", name="x2")
x3 = m1.addVar(vtype="C", name="x3")

m1.setObjective(10*x1 + 14*x2 + 20*x3, sense=GRB.MAXIMIZE)

m1.addConstr(2*x1 + 3*x2 + 4*x3 <= 220, "c0")
m1.addConstr(4*x1 + 2*x2 - x3 <= 385, "c1")
m1.addConstr(x1 + 4*x3 <= 160, "c2")

m1.optimize()

# Create a new model for second LP
m2 = Model("Dual_LP")

y1 = m2.addVar(vtype="C", name="y1")
y2 = m2.addVar(vtype="C", name="y2")
y3 = m2.addVar(vtype="C", name="y3")

m2.setObjective(220*y1 + 385*y2 + 160*y3, sense=GRB.MINIMIZE)

m2.addConstr(2*y1 + 4*y2 + y3 >= 10, "c0")
m2.addConstr(3*y1 + 2*y2 >= 14, "c1")
m2.addConstr(4*y1 - y2 + 4*y3 >= 20, "c2")

m2.optimize()
```

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Here's how the primal/dual solutions line up with the three duality relationships:

- Weak duality says that, for any feasible primal x and dual y , $c^T x \leq b^T y$ (since the primal is a maximization problem and the dual a minimization problem). Specifically at optimality, our solution satisfies this relationship: primal objective = 1100 $\leq$ 1100 = dual objective.
- Strong duality says that if both LPs are feasible and bounded, then: $c^T x^* = b^T y^*$. In our case, objective = dual objective = 1100, so strong duality holds.
- Complementary slackness says that product of each primal slack with its corresponding dual variable is 0, and product of each primal variable with its dual slack (or reduced cost) is 0. In our solutions, all complementary slackness products are zero.

Conclusion: The feasible optimal pair $x^* = (60, 0, 25)$, $y^* = (5, 0, 0)$ satisfies weak duality, achieves strong duality with equal objective value 1100, and meets complementary slackness.

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Question 3 (55 Points). *Suppose that a lumberyard has a supply of 10-ft boards, which are cut into 3-ft, 4-ft, and 5-ft boards according to customer demand. The 10-ft boards can be cut into several sensible patterns, each in such a way that there the leftover material is less than 3-ft. The lumberyard just received an order for 90 3-ft boards, 60 4-ft boards, and 60 5-ft boards.*

1. *Determine all sensible patterns the lumberyard may use to cut the 10-ft boards. [10 pts]*
2. *The lumberyard would like to use as few 10-ft boards as possible in meeting the demand.*
 - (a) *Write out a mathematical formulation for the problem as an integer linear program. [10 pts]*
 - (b) *Solve this problem using Gurobi solver via the Python interface. What is the optimal number of each pattern, and what is the minimum number of boards to cut? [15 pts]*
3. *If it is possible to maintain the minimum number of 10-ft boards to be used, the lumberyard would also like to minimize the amount of scrap pieces leftover (those pieces smaller than 3-ft).*
 - (a) *With respect to the previous solution, model this desire algebraically. [10 pts]*
 - (b) *Modify the model and resolve it. Summarize the results here. [10 pts]*

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Solution 3.1

The six patterns are displayed in the following table:

Pattern	3-ft	4-ft	5-ft
1	3	0	0
2	2	1	0
3	1	0	1
4	0	1	1
5	0	2	0
6	0	0	2

Table 1: Sensible Patterns to Cut Boards

Solution 3.2.a.

Define x_i as the number of 10-ft boards cut into pattern type i , $i = 1, \dots, 6$. Then:

$$\begin{array}{ll} \min & x_1 + x_2 + x_3 + x_4 + x_5 + x_6 \\ \text{subject to} & 3x_1 + 2x_2 + 1x_3 \geq 90 \\ & 1x_2 + 1x_4 + 2x_5 \geq 60 \\ & 1x_3 + 1x_4 + 2x_6 \geq 60 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \in \mathbb{Z}_+ \end{array}$$

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Solution 3.2.b.

```
from gurobipy import *

nWidth = 3
nPattern = 6

widthsInPatterns = [
    [3,2,1,0,0,0],
    [0,1,0,1,2,0],
    [0,0,1,1,0,2],
]
demand = [90,60,60]

model = Model("cuttingstock")

patternsCut = []

for i in range(nPattern):
    patternsCut.append(model.addVar(vtype=GRB.INTEGER, obj=1, lb=0, name="
        pattern%d" % (i+1)))

model.modelSense = GRB.MINIMIZE
model.update()

expr = LinExpr()
for i in range(nWidth):
    expr = 0
    for j in range(nPattern):
        expr += widthsInPatterns[i][j]*patternsCut[j]
    model.addConstr(expr, GRB.GREATER_EQUAL, demand[i], name="sum%d" % (i))
model.update()
model.write('cuttingstock.lp')
model.optimize()

if model.Status == GRB.OPTIMAL:
    print("Minimum # of Rolls = " + str(model.ObjVal))
    for v in model.getVars():
        if v.X > 0:
            print("Cut " + str(v.X) + " of " + v.VarName)
```

The minimal number of 10-ft boards to be cut is 83. This can be done by cutting 2 boards into pattern 1, 42 into pattern 2, 9 into pattern 5, and 30 into pattern 6 (among other ways – there are multiple optimal solutions to this integer program). Incidentally, this solution produces 20 feet of scrap.

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Solution 3.3.a. Two modifications need to be made to the original model - changing the objective to minimize scrap, and adding a constraint that forbids the model from doing any worse than the true minimum number of boards cut (83):

Modification 1.

$$\min \quad x_1 + 2x_3 + x_4 + 2x_5$$

Modification 2.

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 83$$

Solution 3.3.b. The new solution is to cut 46 boards into pattern 2, 14 into pattern 4, and 23 into pattern 6. This solution still adheres to the minimum number of total boards cut (83) – even better, as the new restriction was satisfied! – and reduces the amount of scrap from 20 feet to 14 feet.